River Ice Fine-Grained Segmentation: A GF-2 Satellite Image Dataset and Deep Learning Benchmark

Chenxu Wei, Haoxuan Li, Liang Chen, Haohao Zhou, Omirzhan Taukebayev, Wencong Wu, Amirkhan Temirbayev, Lin Han, Lingyan Ran, Hanlin Yin, Peng Wang, Junrui Liu, Xiuwei Zhang, and Yanning Zhang, *Fellow, IEEE*

***Abstract*—Semantic segmentation of river ice images serves as a critical technological foundation for hydrological monitoring and an ice flood early warning system. Current publicly available river ice datasets predominantly utilize UAV-captured images and ground-based photographic observations. To address the limitations of spatial coverage in existing datasets, we present NWPU_YRCC_GFICE—a satellite remote sensing dataset constructed from multispectral GF-2 satellite images. The dataset innovatively categorizes river ice into six fine-grained classes across freeze–thaw cycles and covers river ice data from the Yellow River (Ningxia-Inner Mongolia section) spanning the past ten years. We further establish a comprehensive deep learning benchmark, which evaluates 33 state-of-the-art segmentation models and two improved segmentation models based on YOLO and SegFormer architectures, separately. Experiments are conducted on the NWPU_YRCC_GFICE dataset and three public river ice datasets (NWPU_YRCC_EX, NWPU_YRCC2, and Alberta river ice segmentation datasets). The proposed models exhibit excellent performance, surpassing the state-of-the-art methods. The presented NWPU_YRCC_GFICE dataset and the benchmark enrich the river ice dataset and favor promoting fine-grained river ice segmentation research from satellite view. Our dataset and code are available at https://github.com/ASGOLabMultisourceCooperationGroup/NWPU_YRCC_GFICE**

***Index Terms*—Fine-grained semantic segmentation, river ice dataset, SegFormer, YOLO.**

Received 15 May 2025; revised 10 July 2025; accepted 26 August 2025. Date of publication 1 September 2025; date of current version 23 September 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFE0213600; in part by the National Natural Science Foundation of China under Grant 62476226 and Grant 61971356; in part by the Natural Science Basic Research Program of Shaanxi under Grant 2024JC-DXWT-07 and Grant 2024JC-YBQN-0719; in part by the National Key Research, Guangdong Basic and Applied Basic Research Foundation, under Grant 2024A1515030186; and in part by the Natural Science Foundation of Ningbo under Grant 2023J262. *(Corresponding authors: Xiuwei Zhang; Junrui Liu; Hanlin Yin.)*

Chenxu Wei, Haoxuan Li, Wencong Wu, Lingyan Ran, Hanlin Yin, Peng Wang, Junrui Liu, Xiuwei Zhang, and Yanning Zhang are with the School of Computer Science, Shaanxi Provincial Key Laboratory of Speech and Image Information Processing, National Engineering Laboratory for Integrated Aerospace-Ground Ocean Big Data Application Technology, Ningbo Institute of Northwestern Polytechnical University, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: weichenxu@mail.nwpu.edu.cn; li_haoxuan@mail.nwpu.edu.cn; wencongwu@mail.nwpu.edu.cn; lran@nwpu.edu.cn; iverlon1987@126.com; peng.wang@nwpu.edu.cn; liu.junrui@nwpu.edu.cn; xwzhang@nwpu.edu.cn; ynzhang@nwpu.edu.cn).

Liang Chen and Haohao Zhou are with the Information Center of Yellow River Conservancy Commission, Zhengzhou 450004, China (e-mail: chlg564@163.com; 2901151727@qq.com).

Omirzhan Taukebayev and Amirkhan Temirbayev are with the Al-Farabi Kazakh National University, Almaty 050040, Republic of Kazakhstan (e-mail: omirzhan.taukebayev@kaznu.edu.kz; amirkhan.temirbayev@kaznu.edu.kz).

Lin Han is with the School of Geography and Tourism, Zhengzhou Normal University, Zhengzhou 450044, China (e-mail: hanlin@zznu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3604644

I. Introduction

RIVER ice represents a dynamic natural phenomenon with significant impacts on both ecological systems and human societies. In regions of the Northern Hemisphere where temperatures drop below 0 °C, river ice forms as water freezes under cold weather conditions [1]. The accumulation of river ice can provoke flooding via ice jams, hinder navigation, and disturb local ecosystems. Due to these potential threats, efficient monitoring of river ice is imperative for ensuring public safety, anticipating disasters, and enhancing resource management [2], [3]. A critical technological foundation of river ice monitoring is river ice segmentation, which involves classifying different elements such as ice, water, and riverbanks at the pixel level in image data. It further aids in distinguishing the various states of river ice. The accurate segmentation is vital for a precise analysis of river ice dynamics, thereby advancing monitoring and management practices.

Semantic segmentation has made significant progress since the advent of fully convolutional networks (FCNs) [4], which laid the foundation for modern segmentation techniques. Building on the FCN framework, models such as DeepLab [5], [6], [7], [8], U-Net [9], U-Net++ [10], HRNet [11], and RefineNet [12] significantly advanced the field. Transformer-based models, such as SegFormer [13] and the Segmentation Transformer (SETR) [14], brought innovation through powerful global context modeling. Models like the BiseNet [15], PIDNet [16], and YOLOv8 [17] have focused on balancing accuracy with processing speed, making them suitable for real-time applications. Additionally, large-scale models like the segment anything model (SAM) [18] and others [19], [20], [21] have enhanced scalability and efficiency, enabling diverse applications.

With the continuous development of semantic segmentation technology, various models have been applied to river ice segmentation. For instance, IceHrNet [22], based on the HRNet

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

[11], has been used for high-resolution river ice analysis, while the ICENet [23], built on the BiseNet [15] framework, targets river ice segmentation. Singh et al. [24] applied CNN-based models like the U-Net [9], SegNet [25], DeepLab [6], and DenseNet [26] to classify river ice into categories such as water, drift ice, and anchor ice. UAV-based datasets, such as the Alberta River Ice Segmentation dataset [24], have played a pivotal role in advancing segmentation, with models like the ICENet [23], ICENetV2 [27], and FastICENet [28] demonstrating success in segmenting ice and water features. Similarly, fixed camera datasets [29], [30] are also widely used for river ice monitoring.

In addition to UAV and camera-based datasets, satellite data have been extensively employed for river ice monitoring. For example, Temimi et al. [31] used NOAA-20 and NPP satellites for near-real-time monitoring of ice in northern U.S. and Canadian basins. Muhammad et al. [32] applied MODIS data from Terra and Aqua satellites to monitor freeze–thaw conditions on the Mackenzie River, while Altena and Kääb [2] combined Sentinel-2 and PROBA-V data for ice movement estimation along the Lena River. Additionally, [33] uses Sentinel-1 for detecting ice jams in Finnish rivers. While these studies focused on large-scale tasks like freeze–thaw detection or assessing the extent of ice cover, fine-grained classification of river ice remains essential. Different types of river ice, including drift ice, shore ice, and flat ice, provide key insights into the river's freezing conditions and help predict ice jam formation, critical for flood forecasting. The absence of large-scale, high-quality annotated datasets for river ice segmentation in satellite images remains a critical bottleneck in advancing river ice monitoring research.

To explore detailed river ice classification, we select the Ningxia-Inner Mongolia section of the Yellow River as our study area, in which the river ice phenomenon is very typical. Compared with UAV-captured images and ground-based photographic observation, satellites provide a larger view. We gather GF-2 satellite images of the studied area spanning ten years and covering the freeze-up to break-up period from November to March and construct a satellite remote sensing dataset, named NWPU_YRCC_GFICE, for fine-grained river ice segmentation. It categorizes river ice into six fine-grained classes across the freeze–thaw cycle. Moreover, we develop a comprehensive deep learning benchmark, which evaluates 35 segmentation models based on NWPU_YRCC_GFICE, including two improved segmentation models based on YOLO and SegFormer architecture, separately. The presented NWPU_YRCC_GFICE dataset and benchmark favor promoting fine-grained river ice segmentation research in a large-scale view. The main contributions are summarized as follows.

1) To advance fine-grained river ice segmentation research, we present a satellite remote sensing dataset, namely NWPU_YRCC_GFICE. All images are captured by the GF-2 satellite in the Ningxia-Inner Mongolia section of the Yellow River, in which the river ice phenomenon is very typical. In total, it contains 28 378 images with a size of 768 × 768 pixels. Pixel-level annotations are performed on the NWPU_YRCC_GFICE dataset, including eight categories covering six fine-grained types of river ice, land, and water. To date, this dataset is the first publicly available satellite remote sensing dataset for fine-grained river ice classification.
2) Based on NWPU_YRCC_GFICE, we establish a comprehensive deep learning benchmark for fine-grained river ice segmentation, evaluating 33 state-of-the-art segmentation models and providing two recommended solutions: ICEYOLO and ICEFormer. This benchmark offers systematic performance comparison and delivers practical recommendations for different application scenarios. ICEYOLO is based on YOLOv8 to take advantage of YOLO's real-time speed and lightweight architecture, providing an optimal solution for real-time monitoring applications. To adapt irregular shape characteristics of river ice, two improvements are adopted, that is, adding attention at different scales in the encoding stage and designing an adaptive keypoint redistribution (AKR) algorithm in the training stage. ICEFormer is constructed on SegFormer architecture and extended by a multiscale fusion, offering the highest accuracy solution for precision-critical applications.
3) Experiments are conducted on the NWPU_YRCC_GFICE dataset and three public river ice datasets (NWPU_YRCC_EX, NWPU_YRCC2, and Alberta river ice segmentation datasets). The proposed models exhibit excellent performance, surpassing the state-of-the-art methods.

II. Related Work

A. Semantic Segmentation

Since the advent of FCNs [4], which eliminates the need for traditional patch-based methods and facilitates the application of deep learning techniques to the segmentation task, the field of semantic segmentation has made great progress and promotion. Following the FCN, several excellent models emerge. The DeepLab series [5], [6], [7] introduce atrous (dilated) convolution, which expands the receptive field without losing resolution, improving feature extraction from larger contextual regions. The U-Net [9] and its evolution U-Net++ [10] advance segmentation by using skip connections and nested skip pathways, which enhance feature propagation and segmentation precision, especially in biomedical and high-resolution imaging. The RefineNet [12] focuses on refining coarse predictions through networks designed for multipath refinement, enhancing the extraction of fine-grained features. The HRNet [11] distinguishes itself by maintaining high-resolution representations throughout the entire network, thereby setting a high standard for preserving spatial information across various segmentation applications. Collectively, these models, with their innovative architectures, have driven the field forward by enhancing feature extraction, contextual understanding, and precise prediction, paving the way for more sophisticated semantic segmentation methods.

With the emergence of Transformer and its outstanding performance in various fields, recent innovations in semantic segmentation have led to the development of Transformer-based models that emphasize modeling global context. Initially

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

designed for image classification, the Vision Transformer (ViT) [34] has been successfully adapted for segmentation by utilizing a self-attention mechanism to capture broad dependencies. This method improves segmentation accuracy by thoroughly understanding the global context. Building on this, the SETR [14] refines the Transformer framework specifically for pixel-level segmentation by replacing typical convolutions with full self-attention, achieving impressive results on complex datasets. More recently, the Swin Transformer [35] has been acknowledged for its use of a shifted window approach in self-attention. This technique effectively balances the extraction of both local and global features while reducing computational demands. In addition, SegFormer [13] introduces an efficient hierarchical Transformer structure that boosts scalability and accuracy, advancing the role of transformers in semantic segmentation. Furthermore, Mixed-Mask Transformer (MMT) [36] introduces a mixed-mask attention mechanism and progressive multiscale learning strategy to address the foreground–background imbalance problem in high-resolution remote sensing scenes, achieving state-of-the-art performance on standard benchmarks such as ISPRS Potsdam and Vaihingen.

Simultaneously, there has been an increasing emphasis on balancing segmentation accuracy and computational efficiency, which is especially important for real-time applications requiring fast processing. Models such as the BiseNet [15], PIDNet [16], and YOLOv8 [17] employ lightweight architectures to decrease processing time with negligible losses in accuracy, rendering them ideal for real-time use in resource-constrained environments. In addition, Cross-Fusion Net (CF-Net) [37] is a lightweight segmentation network that efficiently captures small-scale semantic details through attention-based fusion. These models achieve remarkable segmentation through techniques such as enhanced feature extraction via parallel convolutional pathways and the integration of effective attention mechanisms. Furthermore, larger models like the SAM [18], along with associated frameworks [19], [20], [21], have been designed to prioritize scalability, robustness, and efficiency. [38] proposes a postprocessing framework that directly integrates SAM's raw outputs with the predictions of semantic segmentation models, effectively enhancing mask quality without requiring additional training. These models adeptly handle a wide range of segmentation tasks, from fine-grained pixel-level delineation to comprehensive image processing, delivering excellent outcomes across numerous applications. Their ability to scale and adapt to various environments has become a crucial advancement in semantic segmentation, enabling high-quality segmentation on a larger scale with faster processing speeds.

B. River Ice Segmentation

With the development of remote sensing technologies, satellites, UAVs, and fixed cameras have been adopted to monitor the formation and development of river ice from different views.

Before deep learning, river ice segmentation relied on manual feature extraction, rule-based methods, and traditional machine learning techniques. The SAR image is commonly used, with techniques such as k-means clustering and grouping pixels based on grayscale or texture characteristics, but this method struggles with unbalanced data distributions [39]. To detect river ice breakup dates, Beaton et al. [40] proposed a calibrated threshold method to distinguish open water and not open water (solid ice, cloud, or a combination of the two) and mitigate the impact of cloud interference. Multithreshold techniques [41] are also used for pixel classification, but are still vulnerable to environmental noise [42]. Bharathi and Subashini [43] introduced a texture-based color segmentation method for infrared river ice images, using K-means clustering combined with Gabor filters for texture segmentation. These methods mainly focus on river ice extraction, while their accuracy is limited by noise, lighting, and the complexity of manual feature extraction.

At this stage, satellite data play a crucial role in advancing river ice monitoring. Most approaches adopt satellite images to perform a global analysis of river ice distribution. Data from different sources, such as VIIRS and MODIS, or TanDEM-X and Landsat [44], [45], are applied to improve the segmentation accuracy. MODIS data are used to follow freeze–thaw cycles in rivers such as the Mackenzie River [32]. These studies have largely concentrated on large-scale monitoring activities, such as detecting freeze–thaw cycles and gauging ice cover extents. In addition to satellite observations, ground-based sensing techniques have also been applied to river ice monitoring. Purnell et al. [46] employed GNSS-IR in combination with the SAR image and machine learning to capture river ice breakup processes with high temporal sensitivity.

The rapid development of deep learning has significantly advanced intelligent river ice monitoring. Singh et al. [24] introduced CNN-based segmentation models (e.g., U-Net [9] and SegNet [25]) to classify river ice into water, drift ice, and anchor ice, achieving pixel-level accuracy. Subsequent studies further optimize feature representation for ice segmentation: the ICENet [23] fuses positional- and channel-wise attention to enhance ice boundary delineation, while ICENetv2 [27] incorporates multiscale fusion to address fine-grained ice variations in UAV-captured images. The FastICENet [28] introduces a lightweight architecture with parallel convolutional pathways, achieving real-time inference speeds while maintaining high segmentation accuracy for dynamic ice monitoring. Ansari et al. [47] proposed an IceMaskNet, an instance segmentation model based on the Mask R-CNN with a ResNetV2-50 backbone, aimed at improving segmentation efficiency and suitability for UAV-based river ice monitoring. Fu et al. [48] proposed the CSEU-Net, a novel U-Net variant integrating a ConvNeXt-U backbone and SE attention modules, specifically designed for segmenting river ice floes in UAV-based grayscale images. Zhao et al. [49] developed a transfer learning-based framework integrating channel–spatial attention mechanisms and pyramid pooling to address fine-grained river ice segmentation in high-latitude urban environments. The above algorithms utilize UAV images to segment river ice. Promoted by the fast development of UAV technologies, UAVs with these excellent algorithms provide a fast, convenient, wide-range, and high-resolution way to monitor river ice.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COLLECTION OF RIVER ICE SEGMENTATION DATASETS CATEGORIZED BY THE COLLECTION PLATFORM

#	Dataset	Collection Platform	Volume	Classes	Location	Resolution
1	NWPU_YRCC [23]	UAV	814	3	Yellow River, China	1,600 × 640
2	NWPU_YRCC_EX [28]	UAV	887	3	Yellow River, China	1,600 × 640
3	NWPU_YRCC2 [28]	UAV	1,525	4	Yellow River, China	1,600 × 640
4	Alberta River Ice [24]	UAV+Ground	255	3	Rivers across Alberta, Canada	1,134 × 1,009 ∼ 1,462 × 1,463
5	SHR_RIDS [49]	UAV	1,161	3	Songhuajiang River, China	3,840 × 2,160
6	Ansari et al. [47]	UAV	115	6	Dauphin River, Canada	1,280 × 720 ∼ 5,472 × 3,648
7	Fu et al. [48]	UAV	65	2	Yellow River, China	1,920 × 1,080
8	Zhao et al. [52]	UAV	1,101	3	Songhuajiang River, China	3,840 × 2,160
9	Ansari et al. [50]	Ground	1,795	5	Dauphin River, Canada	–
10	IPC_RI_IDS [30]	Ground	15,600	5	Nenjiang River, China	1,280 × 720
11	Pei et al. [29]	Ground	1,893	4	N. Saskatchewan River, Canada	820 × 500
12	NIMS [51]	Ground	1,829	3	Milwaukee River, U.S.	1,600 × 640 ∼ 1,281 × 1,081
13	NWPU_YRCC_GFICE	Satellite	28,378	8	Yellow River, China	768 × 768

In parallel with UAV-based efforts, fixed ground-based cameras have also been adopted for long-term river ice monitoring using deep learning techniques. Ansari et al. [50] applied a Mask R-CNN-based instance segmentation algorithm to an oblique shore-based image captured along the Dauphin River, enabling effective segmentation and classification of river ice types from camera-collected images. The Rice-Net [51] is presented to segment images from the U.S. Geological Survey fixed river camera network to calculate the fraction of ice coverage, to automatically generate ice flags. It contains a two-stage process: introducing a binary classifier based on the FCN with the Softmax activation function before the segmentation step to filter out ice-free images and then inputting the features of the ice-affected images into a pyramid attention network (PAN) to segment the ice on the water surface. Pei et al. [29] proposed a hybrid pipeline for the North Saskatchewan River ice analysis, including classification, geometric rectification, and segmentation, in which a simple CNN architecture is adopted to classify the image into four classes and a UNet architecture is utilized to segment river ice. These works demonstrate the feasibility and robustness of applying deep learning to fixed-camera images, particularly for long-term and oblique-view monitoring of river ice.

Satellite images have also been increasingly integrated with deep learning methods. Temimi et al. [31] applied the U-Net architecture to classify NOAA-20 and NPP satellite images into seven types, including water, land, vegetation, snow, river ice, cloud, and cloud shadow. Altena et al. [2] used Sentinel-2 and PROBA-V to estimate ice block motion, while Sentinel-1 is utilized for detecting ice jams under cloudy conditions [33]. These methods focus on large-scale dynamic monitoring but lack the granularity required for fine-level classification of specific ice types.

Although significant progress has been made in river ice monitoring, fine-grained river ice segmentation across freeze–thaw cycles remains a critical challenge. This task is especially important for accurately monitoring ice dynamics and forecasting ice jams.

III. NWPU_YRCC_GFICE DATASET

A. Motivation

High-quality river ice datasets play an indispensable role in river ice monitoring based on deep learning. As shown in Table I, the existing public river ice datasets can be folded into three groups according to the data collection platform, namely UAV-based datasets, satellite-based datasets, and ground-camera-based datasets.

Several UAV-based datasets have been developed to advance the intelligent monitoring of river ice. The NWPU_YRCC dataset [23], collected from UAV images over the Ningxia-Inner Mongolia reach of China's Yellow River, includes 814 images annotated into three categories (ice, water, and shore) with a resolution of 1600 × 640 pixels. The NWPU_YRCC_EX dataset [28], an extension of the previous dataset, consists of 887 images, also from the Yellow River, and covers three categories. Similarly, the NWPU_YRCC2 dataset [28] expands to 1525 images with four categories (drift ice, shore ice, water, and shore), addressing finer-grained classification needs for ice management. Singh et al. [24] proposed a comprehensive benchmark on surface ice classification, leveraging high-resolution images and videos captured by UAVs and bridge-mounted cameras over Alberta rivers. The SHR_RIDS dataset [49], comprising 1161 UAV images captured from the middle reaches of the Songhua River near Harbin, China, focuses on high-latitude urban river ice with three pixel-level classes: ice cover, drift ice, and water. The dataset collected from the Dauphin River, presented by Ansari et al. [50], includes 115 images and six categories (frazil pan, broken ice, frazil slush, border ice, ice cover, and open water). The dataset presented by Fu et al. [48] focuses on the Yellow River's Binzhou section. It comprises 65 original UAV images augmented to 27 180 training samples through Gaussian noise, blurring, and rotation, targeting binary segmentation (ice versus water). In addition, the dataset introduced [52] comprises 1101 UAV-captured images from the Songhua River in Northeast China, covering ice channels formed by air-cushion

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

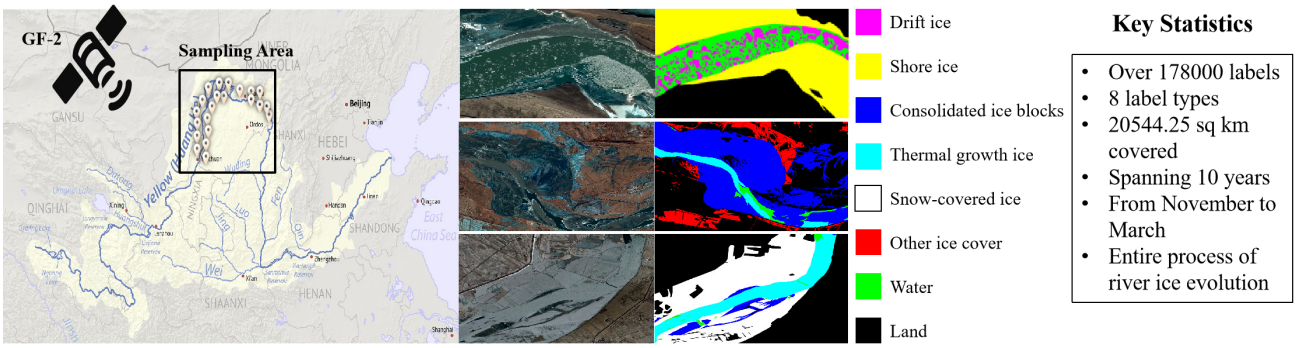


Fig. 1. NWPU_YRCC_GFICE represents an extensive remote sensing dataset obtained via the GF-2 satellite and annotated by hand. The dataset focuses on the redundant classification and segmentation of river ice. It categorizes river ice into six types: drift ice (pink), shore ice (yellow), thermal growth ice (dark blue), consolidated ice blocks (light blue), a mixture of snow with ice or others (white), and other ice cover (red). Additionally, it includes water (green) and land (black).

icebreakers. This dataset provides pixel-level annotations for three categories: ice sheet, ice channel, and background. These datasets, based on UAV-captured images, are invaluable for understanding river ice behavior.

In addition to UAV-based datasets, several ground fixed-camera-based datasets have been introduced. The IPC_RI_IDS dataset [30], comprising 15 600 fixed-camera images captured from the Nenjiang River in China, provides continuous monitoring of ice dynamics across five distinct freeze–thaw stages (ice frozen, break-up initiation, drifting, break-up completion, and ice-free), with pixel-level annotations for ice, water, and background categories. Furthermore, the dataset presented by Ansari et al. [47] includes 1795 oblique shore-based images captured from the Dauphin River in Canada, annotated into five categories: open water, border ice, ice cover, frazil slush, and frazil pan/surface collar ice. The dataset introduced by Pei et al. [29] consists of long-term time-lapse images captured by a public rooftop camera with a distant, oblique view of the North Saskatchewan River in Canada. It contains 1,893 images and spans multiple freeze-up and breakup periods across several years, with manual annotations of ice pans and full ice cover used for segmentation and shape analysis. The NIMS images dataset [51], captured using ground-based cameras along the Milwaukee River in Wisconsin, U.S., is annotated for river ice segmentation into three categories: water, ice, and other. While the ground-mounted fixed-camera approach allows for continuous monitoring, it has limitations in terms of spatial coverage.

Satellite image addresses the limitations of UAVs and stationary cameras by providing extensive spatial coverage, enabling the observation of expansive river ice dynamics. Upon examining existing datasets for semantic segmentation in river ice monitoring, although there are many UAV and ground camera-based datasets, no publicly available satellite remote sensing image dataset for river ice classification and fine-grained semantic segmentation is available. To advance deep learning research in this area, we have created a multispectral satellite image dataset specifically for river ice fine-grained segmentation. This dataset is a collaborative effort between Northwestern Polytechnical University and the Yellow River Conservancy Commission Information Center, and it is designated as the NWPU_YRCC_GFICE River Ice dataset. Here, "NWPU_YRCC" signifies Northwestern Polytechnical University/Yellow River Conservancy Commission, while "GFICE" denotes river ice images captured by the GF-2 satellite. It comprises 28 378 satellite images from the Yellow River in China, spanning multiple freeze–thaw cycles. Totally, it covers eight categories and classifies river ice into six distinct types with fine pixel-level manual annotation.

Our motivation for the NWPU_YRCC_GFICE River Ice dataset is to advance satellite-based river ice segmentation and enable more accurate, fine-grained, and large-scale monitoring of river ice dynamics.

B. Dataset Constructions

1) Data Collection: To automatically monitor river ice dynamics throughout the ice season in a large-scale view, we collect 37 GF-2 satellite multispectral images from 2015 to 2024, covering the months from November to March, including periods of ice formation, growth, and breakup. These images are acquired from the Ningxia-Inner Mongolia section of the Yellow River. The longitude of the covered area is 106.3–111.5, and the latitude is 38.3–40.9. As shown in Fig. 1, it covers the entire "U"-shaped part of the river. We chose this area because the ice phenomenon in this area is the most typical in the Yellow River Basin. The GF-2 satellite data consists of two parts: a four-spectral (red, green, blue, and near infrared) multispectral image with a resolution of 4 m and a panchromatic image with a resolution of 0.8 m. The size of these images ranges from $46\,320 \times 33\,169$ to $39\,392 \times 30\,008$ pixels.

2) Data Preprocessing: Some preprocessing procedures are required to be performed on original satellite images, including panchromatic image orthorectification, multispectral and panchromatic image matching, multispectral image orthorectification, and pan-sharpening of orthorectified panchromatic and multispectral images. After preprocessing, we obtain orthorectified multispectral images with a resolution of 0.8 m, which are used as the input to manual annotation.

3) Manual Annotation: There are several tools for semantic segmentation annotations, such as Image Labeler and LabelMe, which are typically designed to annotate objects with simple shapes, like vehicles. Practical annotation experience revealed challenges with these tools when dealing with satellite images of river ice, which exhibit substantial variability in size

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

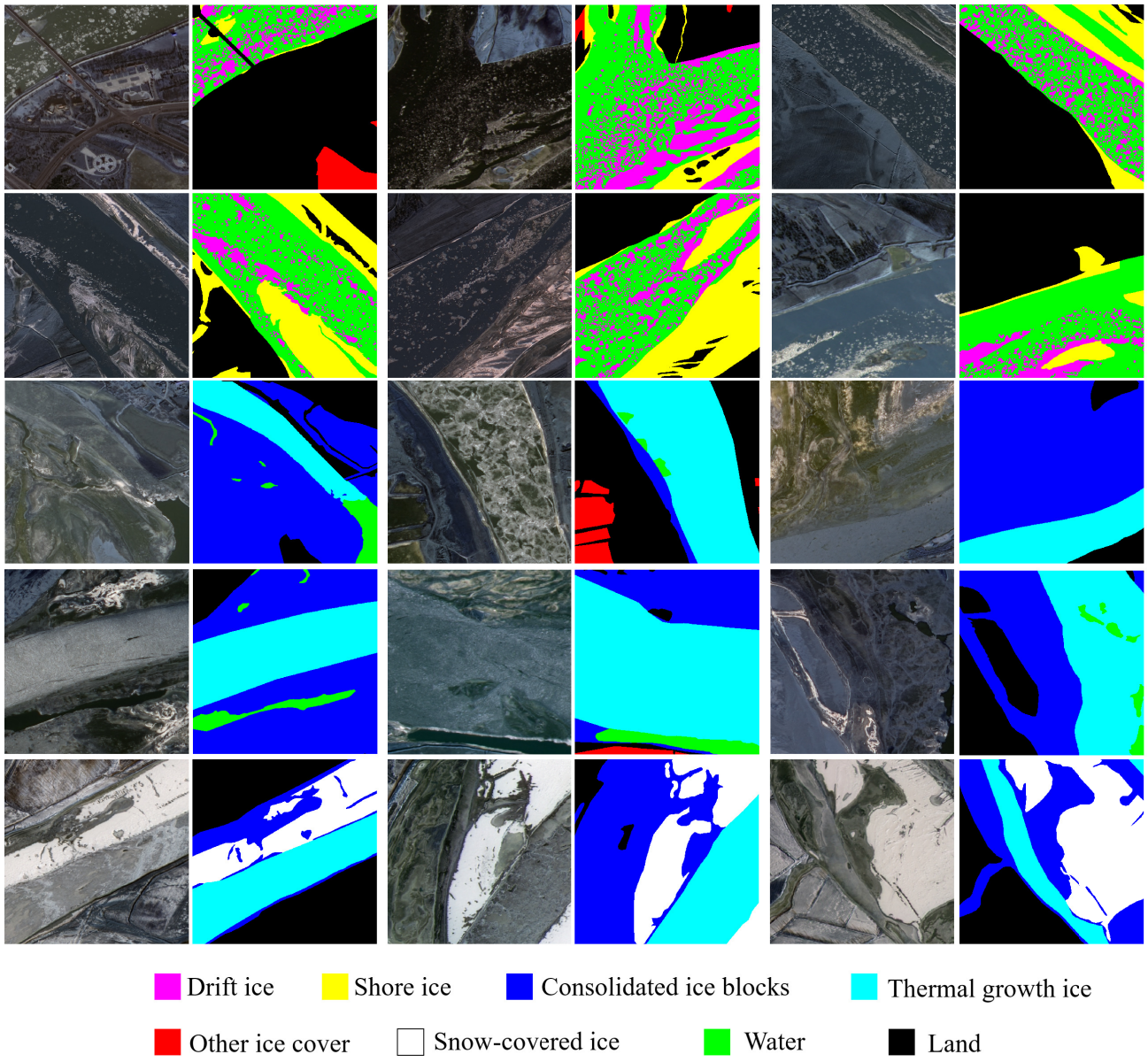


Fig. 2. Visualization of river ice segmentation across various ice types. This figure presents a set of images from the NWPU_YRCC_GFICE dataset, showcasing the segmentation of different river ice categories. Each pair consists of the original image on the left and its corresponding segmentation mask on the right. The categories include drift ice, bank ice, consolidated ice blocks, thermal growth ice, other ice cover, mixture of snow with ice or others, water, and other. The segmentation process is highlighted by distinct color coding for each ice type, providing a clear illustration of the dataset's detailed classification.

and form. Therefore, we adapt the ENVI Classic 5.3 software to produce three-channel pseudo-RGB images. Then, these images are finely annotated in Photoshop, distinguishing eight categories: shore ice, drift ice, consolidated ice blocks, thermal growth ice, mixture of snow with ice or others, water, other ice cover, and land. This process is highly labor-intensive and ultimately completes the annotation of 37 images.

4) Dataset Description: Since the size of each satellite image is approximately 40 000 × 30 000 pixels, it is hard to input the whole satellite image into a semantic segmentation neural network. Therefore, these satellite images are cropped to patches with a size of 768 × 768 pixels, resulting in a collection of 28 378 images.

In our dataset, the river ice images are segmented into three broad categories, that is, open water, river ice, and land. Moreover, according to the characteristics and development of river ice, the river ice is further classified into six fine-grained categories. Fig. 2 shows some typical samples. The detailed description of each category is as follows.

1) Open Water: It refers to areas in the image that are not frozen, representing the water surface or flowing water bodies. Open water exhibits a smoother texture compared to ice.

2) Shore Ice: It is characterized by a smooth, uniform surface, forms along river banks and channel bars. It is also named as border ice or border ice in some literature. As shown in the first two rows of Fig. 2, shore ices are marked in yellow. The shore ice along the bank is usually in the shape of a strip, while the shore ice along the channel bar is usually in the shape of a collar.

3) Drift Ice: It refers to scattered ice floes that form during the floating ice period as a result of temperature drops. These ice float on the water surface and drift with the current.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

Drift ice may undergo different statuses, such as frazil slush, frazil ice pan, and crusty pan, along with temperature change. With drifting on the water surface, they will join/collide with other ice; small ice pans may join together to form a big ice pan/strip. As shown in the first two rows of Fig. 2, this kind of ice (marked by pink) has diverse shapes and sizes. The size ranges from several pixels to thousands of pixels.

4) Thermal Growth Ice: Typically, this type of ice forms in environments where water bodies are still or where the current is exceedingly slow. As the water temperature drops below the freezing point, the surface begins to freeze, resulting in the formation of a flat and continuous ice layer.

5) Consolidated Ice Blocks: This kind of ice forms in environments with rapid water flow, particularly in the fast currents or narrow sections of rivers. For example, drift ice blocks floe with the current. As the temperature decreases, the amount of drift ice increases; they accumulate together so that the drift ice density continuously increases. When they reach the narrow river channel, they are stuck in the river channel and no longer drift. Meanwhile, if the water cools quickly, they will freeze up and never drift to form a continuous large sheet of ice, that is, a consolidated ice block. The morphology of this ice is more chaotic, and it sometimes appears as though the ice blocks have been randomly stacked together.

6) Mixture of Snow With Ice or Others: It refers to the mixture forms when snow and ice layers or others in the river channel combine after a snowfall. It exhibits higher brightness and reflectivity.

7) Other Ice Cover: It refers to ice that forms on land, typically near wetlands or adjacent areas to the river, but not directly connected to the river channel. This type of ice may be formed by the freezing of water overflowing from the rising water level in the river channel.

8) Land: It refers to the other ground areas outside the river channel and not covered by ice.

C. Dataset Characteristics

To our best knowledge, the presented NWPU_YRCC_GFICE dataset is the first satellite image dataset for fine-grained river ice semantic segmentation. All images are captured by the GF-2 satellite in the Ningxia-Inner Mongolia section of the Yellow River, in which the river ice phenomenon is very typical. In total, it contains 28 378 images with a size of 768 × 768 pixels. Compared with the existing river ice datasets with semantic labels, it is currently the largest one. To fine explore the river ice start, development, and evolution from satellite view, we carefully classify the river ice into six fine-grained subclasses at the pixel level. Only the dataset collected from the Dauphin River, presented by Ansari et al. [50], can be on par with our dataset in terms of the diversity of river ice types. Meanwhile, our dataset spans river ice conditions of the Yellow River (2015–2024) during freeze–thaw seasons, with image acquisitions concentrated between November and March, and the comparable dataset only contains 115 labeled images collected from November to March of three consecutive years (2017 to 2020). Therefore, our dataset has significant advantages in terms of data diversity. In conclusion, the NWPU_YRCC_GFICE dataset enriches the river ice dataset, enabling fine-grained large-scale monitoring of river ice from satellite view.

IV. Method

To establish a comprehensive benchmark for fine-grained river ice segmentation, we propose two baseline models: ICEYOLO and ICEFormer, as shown in Fig. 3. ICEYOLO is a CNN-based model built upon the YOLOv8 architecture, designed for real-time instance segmentation of river ice. In contrast, ICEFormer is a Transformer-based semantic segmentation model, featuring a hierarchical encoder–decoder structure to capture rich spatial and spectral information. In Sections IV-A and IV-B, we detail the architectural designs and key components of both models.

A. ICEYOLO

Since river ice monitoring tasks often require both high segmentation accuracy and real-time inference speed, ICEYOLO leverages the YOLOv8 framework and two improvements to balance precision and efficiency.

Although YOLOv8 performs well on many panoramic instance segmentation tasks with a competitive inference speed advantage, its direct use is not suitable for river ice semantic segmentation. This is mainly attributed to two aspects as follows.

1) The irregular shapes and fragmented boundaries commonly observed in river ice formations make standard instance label conversion prone to geometric distortion. As shown in the third column of Fig. 4, the instance boundary generated by the original YOLOv8, marked by a red rectangle, is incomplete or inaccurate, especially for instances on the image edge and small instances.
2) River ice often exhibits subtle textural variations across different ice types, such as shore ice, drift ice, and consolidated ice blocks.

To address the challenges posed by river ice segmentation, two targeted improvements are introduced as follows.

1) The AKR algorithm is proposed, ensuring more uniform and accurate boundary point distribution during instance generation.
2) The channel attention mechanism (CAM) [53] is incorporated into the neck part of the YOLOv8 architecture, allowing the network to dynamically prioritize informative feature channels and improve segmentation accuracy under complex environmental conditions. The details of the entire model architecture and two improvements are introduced one by one as follows.

1) Model Architecture: Similar to YOLOv8, the ICEYOLO model follows a four-stage architecture: preprocessing, backbone, neck, and segmentation head, as shown in Fig. 3(a). In the preprocessing stage, semantic segmentation masks are transformed into YOLO-compatible instance segmentation labels. This involves the use of the Teh-Chin chain approximation algorithm to extract boundary points, followed by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

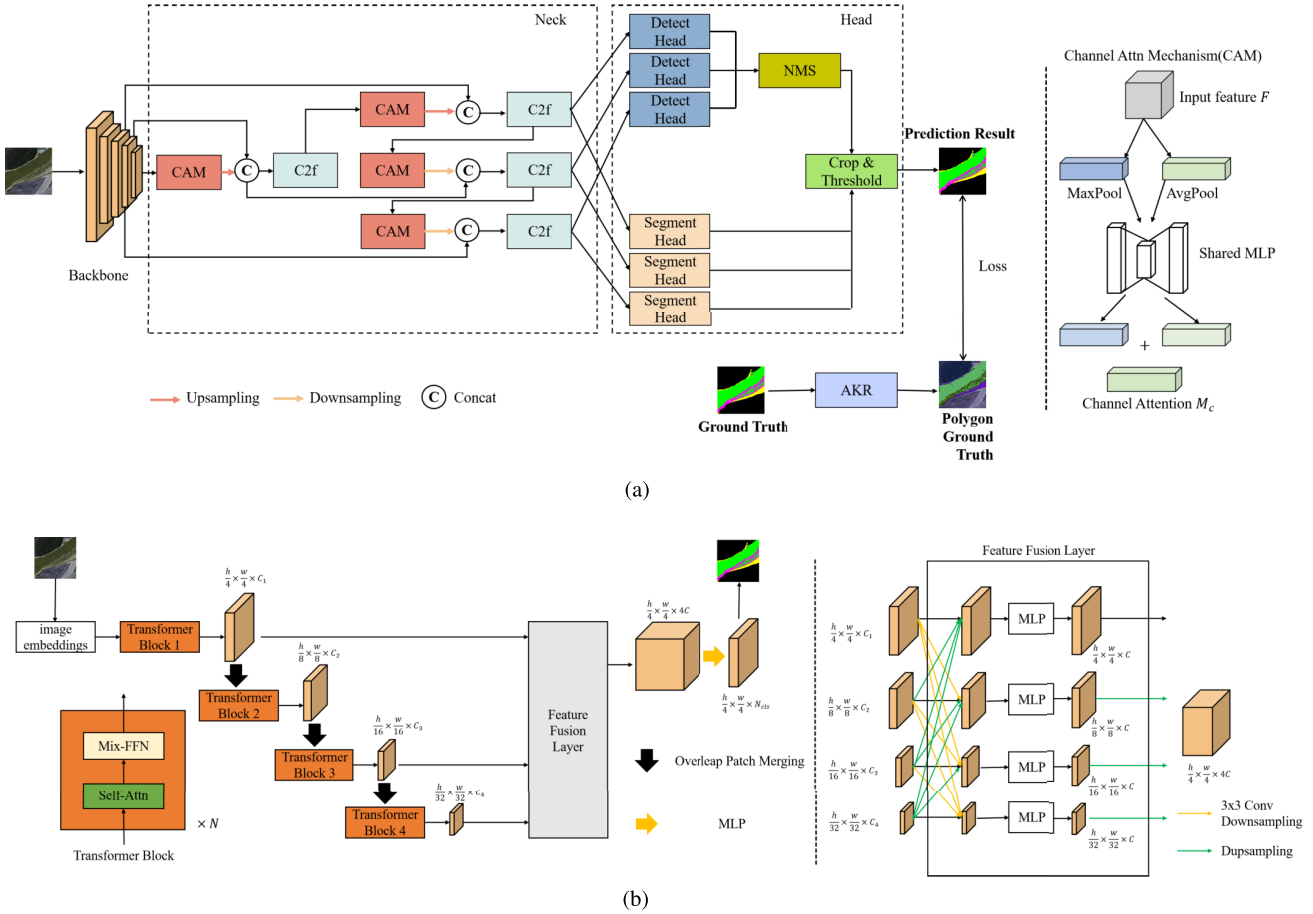


Fig. 3. Architectures of the proposed river ice segmentation models. (a) ICEYOLO, an enhanced YOLO-based river ice segmentation framework. (b) ICEFormer, a Transformer-based river ice segmentation network.

the AKR algorithm to optimize keypoint distribution, preserving boundary integrity and minimizing information loss. The backbone adopts the YOLOv8 feature pyramid design, which incorporates C2f modules and depthwise separable convolutions to efficiently extract multiscale hierarchical features from the input images. Feature maps are then aggregated in the neck module, where the CAM is inserted before each C2f block in the multiscale feature aggregation path of the neck to selectively enhance critical feature representations. Finally, the segmentation head predicts instance masks, bounding boxes, and confidence scores for detected river ice objects.

2) AKR Algorithm: During the transformation from semantic masks to instance labels, the uneven distribution of boundary points can severely distort the geometry of complex river ice structures, thereby impairing segmentation performance. This unevenness often arises when long or intricately shaped river ice objects produce mask contours where some segments are sparsely sampled while others are overly dense, leading to jagged or biased polygonal approximations. The original transformation algorithm in YOLOv8 faces this issue when coping with river ice, as new keypoints are added without considering the overall shape and distribution of the boundaries.

To overcome this, the AKR algorithm is designed to iteratively balance the distribution of keypoints along object boundaries. The algorithm first decomposes the boundary into segments and inserts them into a max-heap sorted by length. The longest segment is selected at each iteration for keypoint insertion, progressively achieving a uniform point distribution. This procedure continues until the target number of keypoints is reached. Finally, keypoints are collected and sorted to reconstruct a smooth and consistent boundary contour. Multithreaded processing is adopted to accelerate computation. The procedural details are outlined in Algorithm 1.

3) Channel Attention Mechanism: The intricate textures and subtle interclass variations characteristic of river ice present challenges for traditional convolutional feature extraction. To address this, a CAM [53] is embedded into the neck module of ICEYOLO. Specifically, CAM modules are inserted before each C2f block in the multiscale feature aggregation path of the neck, enhancing channel-wise feature representations at three different scales. By dynamically adjusting feature weights, CAM assigns higher attention scores to informative channels while suppressing redundant responses. This improves the model's sensitivity to ice-related patterns such as thin drift ice and fragmented shore ice, ultimately boosting segmentation accuracy in complex river scenes. The structure of CAM is depicted by the right part of Fig. 3(a).

B. ICEFormer

ICEFormer is a SegFormer-based semantic segmentation benchmark designed for high-precision classification of river ice. Compared to CNN-based models, the Transformer architecture offers stronger global context modeling, making it particularly suitable for capturing the complex spatial distribu-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

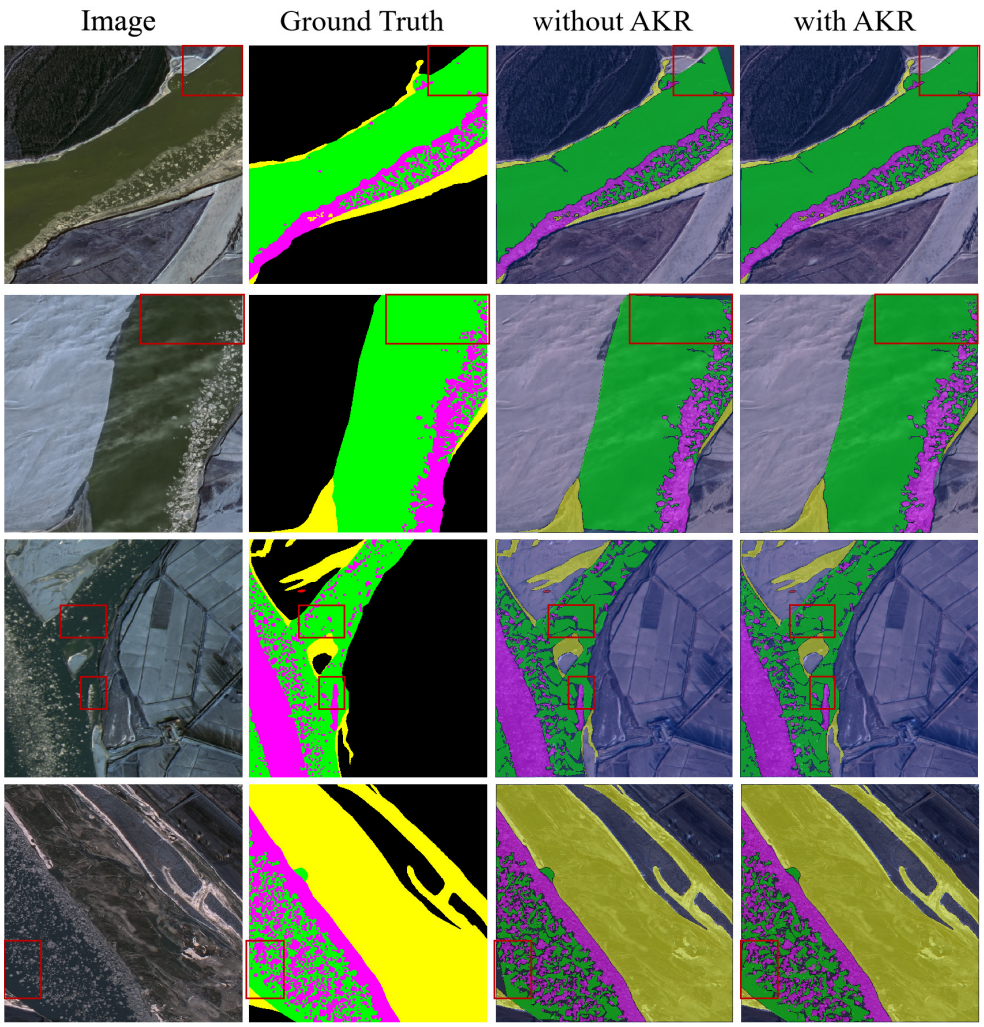


Fig. 4. AKR algorithm. The original algorithm for adding keypoints in YOLOv8 often results in significant loss of important regions, as new keypoints are added without considering the overall shape and distribution of the boundaries. This can lead to poor representation of the river ice contours. In contrast, the proposed AKR algorithm ensures a more balanced and accurate distribution of keypoints along the boundary, preserving the integrity of the original shape and minimizing the loss of crucial details.

tions of different river ice types over large areas. Considering the multiscale characteristics of river ice, that is, the difference in size of diverse river ice instances is huge, we improve the original SegFormer by a well-designed multiscale feature fusion module. The details of the ICEFormer architecture and feature fusion module are introduced as follows.

1) Model Architecture: ICEFormer adopts a hierarchical encoder–decoder architecture, the same as SegFormer. The encoder directly processes the input remote sensing image through multiple Transformer stages, progressively downsampling the feature maps to 1/4, 1/8, and 1/16 resolution of the original input image. Each Transformer block integrates a multihead self-attention mechanism to capture long-range dependencies and a mix feed-forward network (Mix FFN) to enhance local feature extraction via depthwise separable convolutions. This design enables ICEFormer to simultaneously model fine-grained ice features and large-scale spatial structures, addressing the characteristics of river ice scenes.

The decoder aggregates multiscale features by the designed feature fusion module and progressively reconstructs high-resolution segmentation maps. Finally, a multilayer perceptron (MLP) layer is utilized to refine the fused features, generating the output segmentation map with improved spatial consistency and class discrimination.

2) Feature Fusion Layer: To fully exploit multiscale semantic information, a dedicated feature fusion layer (FFL) is introduced before final prediction. As illustrated in the right part of Fig. 3(b), there are totally four groups of feature maps with different scales, produced by our encoder. For each scale, features from other scales are processed through a resolution-aligned process: those from coarser scales are downsampled using 3×3 convolutions, and those from finer scales are upsampled; then, these resized feature maps are aggregated with themselves by a concatenation operation. To minimize interpolation artifacts and preserve fine boundary details, data-dependent upsampling (DUpsampling) [54] is employed for upsampling during resolution restoration. Subsequently, the channel dimensions across all aggregated features of different scales are unified via MLPs, followed by concatenation along the channel axis to produce the fused features.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Polygon Boundary Keypoint Alignment

Require: segments (original boundary points of the polygon), n (target number of keypoints)
Ensure: points (aligned boundary keypoints)

Initialize a max-heap *heap*
total_point_count = len(*segments*) - 1
Initialize the heap
for each pair of consecutive points in *segments* **do**
    length = distance between (*point1.x, point1.y*) and (*point2.x, point2.y*)
    line = create segment object:
    line.index = current segment index
    line.point1 = (*point1.x, point1.y*)
    line.point2 = (*point2.x, point2.y*)
    line.point_count = 1
    line.length = *length*
    push *line* into *heap*
end for
Add points
while *total_point_count* < n **do**
    line = extract max-length segment from *heap*
    line.point_count+ = 1
    update *line* and push back into *heap*
    total_point_count+ = 1
end while
Sort segments by original index
sorted_lines = sort segments in *heap* by original index
Initialize *points* array
Collect points
for each *line* in *sorted_lines* **do**
    add *line.point1* to *points*
    for i from 0 to *line.point_count* - 1 **do**
        $time = i/(line.point_count - 1)$
        $new_x = line.point1.x + time \times (line.point2.x - line.point1.x)$
        $new_y = line.point1.y + time \times (line.point2.y - line.point1.y)$
        add (new_x, new_y) to *points*
    end for
end for
Handle the last remaining point
add the last point of the original segments to *points*
return *points*

V. Experiments

A. Dataset

In addition to NWPU_YRCC_GFICE, we conduct experiments on three publicly available river ice segmentation datasets, that is, NWPU_YRCC_EX, NWPU_YRCC2, and Alberta River Ice Segmentation datasets, to assess the generalization capability of our models. For fair comparison, ICEYOLO and ICEFormer are trained and evaluated individually on each dataset. NWPU_YRCC_EX contains 887 pixel-wise annotated UAV images, splitting into 524 training images, 180 validation images, and 183 test images. The NWPU_YRCC2 dataset comprises 1525 annotated images and follows a 3:1:1 split, resulting in 915 images for training, 305 for validation, and 305 for testing. The Alberta River Ice Segmentation dataset is generated by cropping and augmenting 50 manually annotated UAV and camera-captured images, resulting in 554 training images and 138 testing images.

B. Evaluation Criteria

In our experiments, we adopt two key metrics: mean intersection over union (mIoU) to assess segmentation accuracy, and frames per second (FPS) to reflect inference efficiency. These two metrics together provide a comprehensive evaluation of each model's balance between precision and speed, which is particularly important for real-time monitoring and large-scale satellite image analysis.

C. Experimental Settings

To ensure a fair and rigorous comparison, all models in our benchmark are trained individually with hyperparameter tuning tailored to each architecture. Segmentation heads are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

TABLE II
BENCHMARK ON THE NWPU_YRCC_GFICE DATASET. CNN-BASED MODELS AND TRANSFORMER-BASED MODELS ARE SEPARATED FOR COMPARISON

Method	IoU(%)								mIoU(%)	FPS
	Thermal Growth Ice	Consolidated Ice Blocks	Shore Ice	Drift Ice	Mixture of snow with ice or others	Water	Other Ice Cover	Land		
CNN-based Models										
FCN [4]	55.00	57.42	58.83	57.42	70.11	81.96	93.28	33.26	65.60	25.92
UNet [9]	64.12	65.56	49.18	61.09	58.15	78.29	65.30	89.96	66.45	22.39
DeepLabV3 [7]	75.03	67.29	62.17	56.75	65.61	82.34	76.70	94.95	72.61	20.52
PSPNet [55]	68.63	55.50	46.88	55.22	25.41	78.77	70.01	91.57	61.50	27.02
SegNet [25]	39.66	31.19	1.00	12.89	11.27	59.66	43.08	85.16	35.49	20.98
ERFNet [56]	29.32	44.12	69.59	44.12	67.50	79.75	91.93	14.20	55.31	92.58
RefineNet [12]	43.18	41.94	16.37	28.10	18.93	68.24	55.20	88.23	58.75	18.10
UNet++ [10]	57.02	49.94	2.98	19.25	15.53	72.88	60.21	88.53	45.79	19.81
DeepLabV3+ [8]	72.73	67.20	54.35	58.71	44.33	83.84	76.89	94.71	69.09	24.22
PSANet [57]	62.66	46.62	46.73	54.43	15.55	80.11	68.56	89.77	58.05	19.95
DenseASPP [58]	74.80	64.80	56.21	54.80	40.01	83.53	76.74	94.24	68.14	55.00
BiseNet [15]	56.89	60.08	72.31	60.08	78.81	86.30	94.67	55.43	72.95	148.62
ENCNet [59]	51.73	64.48	57.62	51.89	72.91	82.24	93.99	19.40	66.38	27.31
HRNet [60]	53.53	56.11	77.35	56.11	72.53	86.31	93.60	30.42	66.63	22.27
CCNet [61]	76.68	64.55	64.97	62.94	40.07	85.59	78.46	93.23	70.81	24.09
SemFPN [62]	67.41	39.99	45.36	59.51	64.12	78.52	72.73	94.59	65.28	70.39
APCNet [63]	81.06	65.13	69.26	73.21	50.42	86.12	81.75	94.75	75.21	23.11
ANN [64]	85.54	66.33	79.11	83.57	74.63	87.67	84.68	95.99	82.19	25.80
DANet [65]	84.30	65.70	76.12	79.64	55.99	88.23	83.78	95.39	78.64	21.70
OCRNet [66]	77.78	68.69	66.47	60.73	60.86	84.74	78.77	94.92	74.12	30.66
DDRNet [67]	59.33	42.79	40.01	46.51	37.36	65.59	66.46	91.19	56.15	109.61
KNet [68]	75.11	61.29	65.36	64.82	44.48	86.64	79.46	94.52	71.46	24.89
BiseNetV2 [69]	51.86	32.74	19.30	16.81	16.75	36.14	49.93	82.90	38.30	129.39
STDC [70]	73.40	63.27	58.74	49.45	49.27	77.57	68.49	92.91	66.64	122.47
SegNext [71]	84.43	64.89	75.46	83.50	80.49	87.28	83.21	96.03	81.91	63.79
PIDNet [16]	66.70	56.70	53.59	58.25	53.31	70.11	60.20	84.90	62.97	88.08
ICEYOLO (Ours)	83.66	82.45	76.85	69.19	78.52	86.70	85.88	96.65	82.48	39.21
Transformer-based Models										
ViT [34]	79.95	56.78	72.93	76.51	77.17	85.95	79.65	95.54	78.06	8.94
SETR [14]	53.64	41.29	30.64	28.89	49.89	62.71	61.76	91.49	52.54	5.87
DPT [72]	74.41	71.71	51.31	50.05	64.99	83.35	77.28	95.32	71.05	12.23
Segmenter [73]	61.49	49.63	27.20	19.02	52.47	65.85	65.98	93.38	54.38	14.89
SegFormer [13]	84.22	84.64	78.19	71.96	80.75	89.02	82.71	96.10	83.45	16.71
Swin [35]	77.19	70.83	80.28	70.83	83.89	88.74	95.88	79.25	82.31	20.20
TWINS [74]	71.89	66.52	77.78	66.52	80.97	88.40	95.81	75.17	79.68	38.24
ICEFormer (Ours)	84.23	85.69	78.21	72.34	82.04	89.30	82.72	96.10	83.83	15.57

adapted to match the number of target classes in each dataset. All input images are resized to a resolution of 768 × 768 pixels. Segmentation heads are modified as necessary to accommodate the class numbers specific to each dataset. All hyperparameters have been systematically optimized through validation experiments to achieve optimal performance for each model architecture.

1) ICEYOLO Training Settings: ICEYOLO is optimized using the Adam optimizer with an initial learning rate of 0.01. The learning rate follows a linear decay schedule over 500 training epochs. A batch size of 40 is used during training. Random image flipping is applied for data augmentation. The AKR algorithm is applied to each polygon instance, with the target number of keypoints set to 1000, which was experimentally determined to be optimal.

2) ICEFormer Training Settings: ICEFormer is trained using the AdamW optimizer, with an initial learning rate of $6e^{-5}$ and a weight decay of 0.01. A polynomial learning rate decay policy is adopted. Training is conducted for 85 epochs, with a batch size of 12. Data augmentation includes random scaling within the range [0.5, 2.0] and random horizontal flipping.

3) Hardware Environment: All experiments are implemented using PyTorch and conducted on four NVIDIA GeForce RTX 4090 GPUs. Inference speed FPS is measured separately on a single NVIDIA RTX 3090 GPU for consistency across all models.

4) Loss Functions: ICEYOLO adopts the original segmentation loss function of YOLO, and ICEFormer inherits the pixel-wise cross-entropy loss from SegFormer.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

TABLE III

PERFORMANCE COMPARISON ON THREE PUBLIC RIVER ICE SEGMENTATION DATASETS: NWPU_YRCC_EX, NWPU_YRCC2, AND THE ALBERTA RIVER ICE SEGMENTATION DATASET. OUR PROPOSED MODELS (ICEYOLO AND ICEFORMER) ARE HIGHLIGHTED IN BOLD. THE BEST-PERFORMING RESULTS FOR EACH DATASET ARE MARKED IN RED, WHILE THE SECOND-BEST RESULTS ARE MARKED IN BLUE

Method	mIoU (%)		
	NWPU_YRCC_EX [29]	**NWPU_YRCC2** [29]	**Alberta River Ice Segmentation dataset** [24]
ENet [75]	89.94	81.37	80.75
CGNet [76]	90.93	80.49	81.62
DABNet [77]	90.05	80.12	82.13
FPENet [78]	83.61	76.89	79.89
FSSNet [79]	89.15	79.63	78.04
LEDNet [80]	91.27	81.81	81.56
ContextNet [81]	87.16	77.83	78.64
FastSCNN [82]	87.13	77.36	78.11
ERFNet [56]	91.57	81.58	81.94
LinkNet [83]	90.45	81.29	82.12
BiSeNet [15]	89.72	80.09	81.24
STDC-2 [70]	91.18	81.46	81.86
PP-LiteSeg-T2 [84]	89.54	79.99	80.39
PIDNet-S [16]	90.21	80.29	81.27
PIDNet-L [16]	91.25	81.22	82.10
ICEYOLO (Ours)	88.65	92.34	93.17
ICEFormer (Ours)	92.98	90.89	96.82

D. Benchmark on the NWPU_YRCC_GFICE Dataset

Table II presents the benchmarking results of 33 existing models and the two proposed models, ICEYOLO and ICEFormer, on the NWPU_YRCC_GFICE dataset. Models are categorized into CNN-based and Transformer-based groups for a more structured comparison.

Among CNN-based methods, DeepLabV3+, OCRNet, and SegNext achieve relatively high mIoU scores, demonstrating strong segmentation capabilities. ICEYOLO, based on the YOLOv8 architecture and improved with AKR and multi-scale attention mechanisms, achieves a superior mIoU of 82.48% while maintaining a high inference speed of 39.21 FPS. This balance of accuracy and efficiency highlights the advantages of ICEYOLO for practical, river ice monitoring applications. Conversely, the SegNet exhibits the poorest performance among CNN-based methods with only 35.49% mIoU. The SegNet's poor performance is attributed to its reliance on max-pooling indices transmission and MaxUnpool2d operations, leading to significant spatial detail loss and limited boundary reconstruction capability for the complex spectral and geometric patterns in satellite-based river ice imagery.

For Transformer-based methods, the SegFormer, Swin Transformer, and ViT exhibit competitive segmentation performance, leveraging their strong global context modeling capabilities. Notably, the proposed ICEFormer achieves an mIoU of 83.83%, outperforming SegFormer and other Transformer baselines. ICEFormer also maintains reasonable inference efficiency with 15.57 FPS, validating the effectiveness of the multiscale feature fusion and hierarchical enhancements tailored for fine-grained river ice segmentation. In contrast, SETR exhibits the poorest performance among Transformer-based methods with only 52.54% mIoU. SETR's suboptimal performance stems from the absence of multiscale feature fusion mechanisms and large patch usage, which results in fine-grained detail loss and blurred boundaries.

In summary, ICEFormer achieves superior segmentation accuracy, outperforming all baseline methods, whereas ICEYOLO maintains a favorable balance between segmentation accuracy and real-time inference efficiency. The results demonstrate that the architectural adaptations and preprocessing strategies introduced in this study effectively address the challenges posed by temporal variability and morphological complexity in satellite-based river ice segmentation.

E. Comparison Experiments

To assess model generalization, we further evaluate ICEYOLO and ICEFormer on three public datasets: NWPU_YRCC_EX, NWPU_YRCC2, and Alberta. As shown in Table III, both models outperform all other 14 segmentation models across all datasets.

ICEYOLO achieves the top MIoU score on NWPU_YRCC2 (92.34%), while ICEFormer leads on NWPU_YRCC_EX with the MIoU score 92.98% and on the Alberta River Ice Segmentation dataset with the MIoU score 96.82%. These results demonstrate that our models perform well on other river ice datasets.

F. Ablation Studies

To validate the effectiveness of individual components in our proposed models, we conduct comprehensive ablation studies on the NWPU_ YRCC_ GFICE dataset. These experiments systematically evaluate the contribution of each proposed module by progressively adding components to the baseline architectures. Table IV presents the quantitative results of this ablation study.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

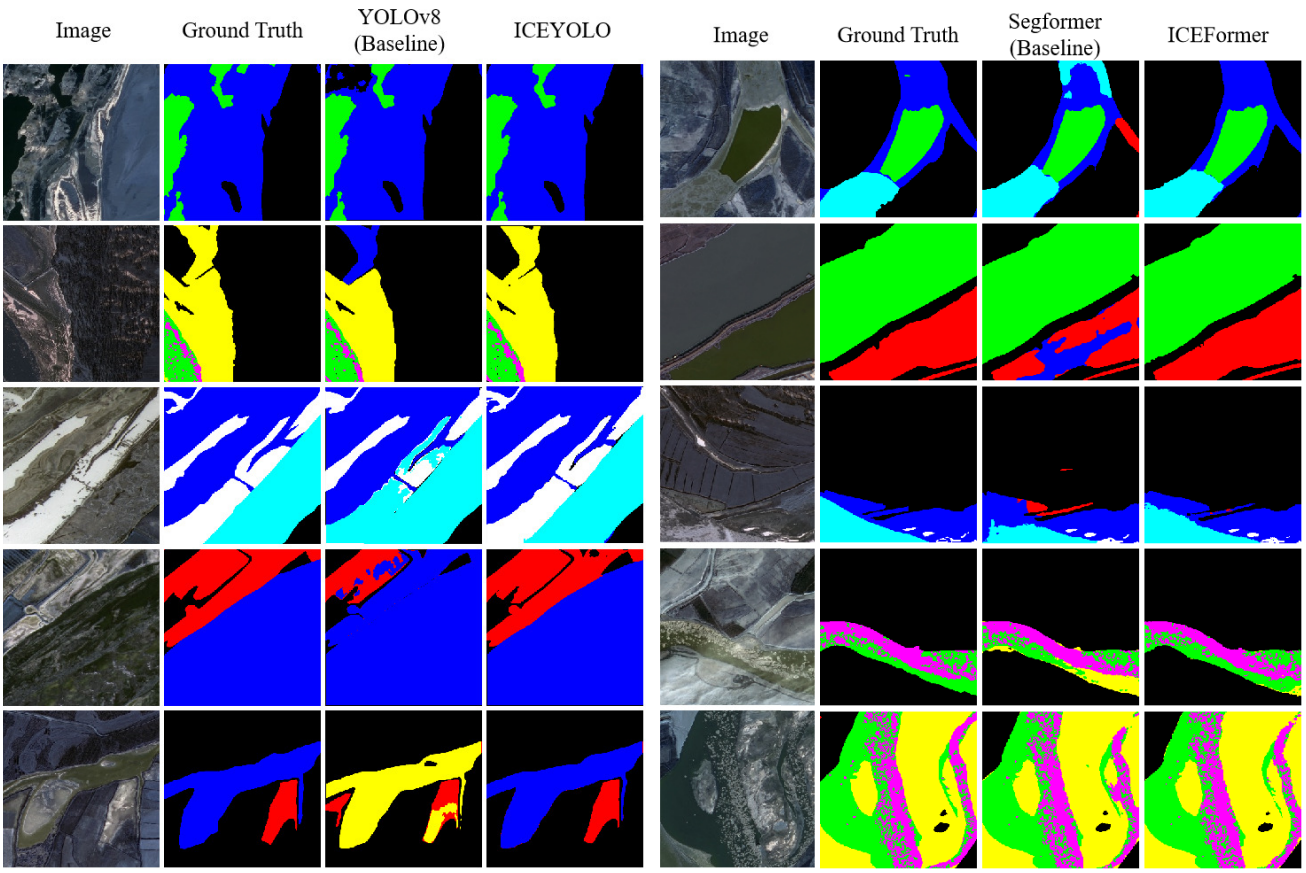


Fig. 5. (Left) Visual comparison of YOLOv8 versus ICEYOLO and (right) SegFormer versus ICEFormer on the NWPU_YRCC_GFICE dataset.

TABLE IV
ABLATION STUDY RESULTS ON THE NWPU_YRCC_GFICE DATASET

Method	mIoU (%)
ICEYOLO Variants	
Baseline (YOLOv8)	80.40
+ AKR	81.08
+ CAM	82.10
+ AKR + CAM (ICEYOLO)	82.48
ICEFormer Variants	
Baseline (SegFormer)	83.45
+ FFL (ICEFormer)	83.83

AKR: Adaptive Keypoint Redistribution;
CAM: Channel Attention Mechanism;
FFL: Feature Fusion Layer

1) ICEYOLO Component Analysis: For ICEYOLO, we evaluate the individual and combined contributions of the AKR algorithm and CAM. The baseline configuration uses the original YOLOv8 architecture without any modifications. Fig. 5 (left) visualizes the comparative results between YOLOv8 and ICEYOLO.

The baseline achieves an mIoU of 80.4%. The incorporation of AKR alone yields a performance improvement of 0.68%, reaching 81.08% mIoU. This improvement demonstrates that the proposed keypoint redistribution strategy effectively preserves boundary integrity during the semantic-to-instance label conversion process, particularly benefiting the segmentation of irregularly shaped river ice formations.

The incorporation of CAM alone elevates the mIoU to 82.1%, reflecting a notable improvement of 1.7% over the baseline. This significant enhancement suggests that the CAM effectively captures subtle textural differences among various ice types, allowing the model to classify pixels more accurately based on their contextual information. Furthermore, the AKR module enhances segmentation precision for complex and irregularly shaped ice formations. When both AKR and CAM are integrated into the full ICEYOLO framework, the model attains its highest performance, reaching 82.48% mIoU. The superiority of this combined approach is further illustrated in the qualitative comparisons shown in Fig. 5 (left), highlighting the improved delineation and class-specific accuracy achieved by the proposed modules relative to the baseline.

2) ICEFormer Feature Fusion Analysis: For ICEFormer, we evaluate the contribution of the proposed FFL by comparing the enhanced model against the standard SegFormer baseline. The baseline SegFormer architecture achieves 83.45% mIoU on our dataset, establishing a strong foundation for Transformer-based segmentation. Fig. 5 (right) visualizes the comparative results between SegFormer and ICEFomer.

The incorporation of the multiscale FFL increases the performance to 83.83% mIoU, representing an improvement of 0.38%. While this enhancement appears modest in absolute terms, it is statistically significant and demonstrates the value of explicitly modeling multiscale feature relationships for fine-grained river ice classification. By fusing features across different dimensions, the FFL substantially improves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

the model's capacity to segment classes with higher accuracy, effectively capturing ice formations of various scales from small drift ice fragments to large consolidated blocks. This improvement is further corroborated by the qualitative results presented in Fig. 5 (right), which illustrate the enhanced discriminative capability enabled by this fusion strategy.

VI. Conclusion

In this study, we address the critical gap in fine-grained river ice segmentation using satellite images by introducing the NWPU_YRCC_GFICE dataset. This dataset, constructed from multispectral GF-2 satellite images, provides pixel-level annotations across eight detailed categories and spans multiple freeze–thaw cycles of the Yellow River. It is the first publicly available satellite-based dataset specifically designed for fine-grained river ice classification.

To benchmark segmentation performance, we develop two tailored models: ICEYOLO and ICEFormer. ICEYOLO, based on the YOLOv8 architecture with targeted enhancements including AKR and CAM, achieves an excellent balance between segmentation accuracy and inference speed. ICEFormer, built upon the SegFormer framework with multiscale feature fusion, achieves the highest segmentation accuracy among all evaluated methods. Extensive experiments demonstrate that both models outperform numerous existing methods on the NWPU_YRCC_GFICE dataset, while maintaining strong generalization capability across three public river ice datasets.

The NWPU_YRCC_GFICE dataset and the proposed models provide benchmarks for advancing satellite-based river ice monitoring. Future work will explore more efficient Transformer architectures and self-supervised learning strategies to further enhance segmentation performance under limited labeled data conditions.

References

[1] I. Zakharov, T. Puestow, A. A. Khan, R. Briggs, and P. Barrette, "Review of river ice observation and data analysis technologies," *Hydrology*, vol. 11, no. 8, p. 126, Aug. 2024.

[2] B. Altena and A. Kääb, "Quantifying river ice movement through a combination of European satellite monitoring services," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 98, Jun. 2021, Art. no. 102315.

[3] A. Sharifi, "Development of a method for flood detection based on Sentinel-1 images and classifier algorithms," *Water Environ. J.*, vol. 35, no. 3, pp. 924–929, Aug. 2021.

[4] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[5] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.

[6] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, Atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[7] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking Atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[8] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[9] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351, Munich, Germany. Cham, Switzerland: Springer, 2015, pp. 234–241.

[10] Z. Zhou, M. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.*, vol. 11045, 2018, pp. 3–11.

[11] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.

[12] G. Lin, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for high-resolution semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1925–1934.

[13] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

[14] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.

[15] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "BiseNet: Bilateral segmentation network for real-time semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 325–341.

[16] J. Xu, Z. Xiong, and S. P. Bhattacharyya, "PIDNet: A real-time semantic segmentation network inspired by PID controllers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 19529–19539.

[17] G. Jocher, A. Chaurasia, and J. Qiu. (2023). *Ultralytics YOLOv8*. [Online]. Available: https://github.com/ultralytics/ultralytics

[18] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[19] X. Zhao et al., "Fast segment anything," 2023, *arXiv:2306.12156*.

[20] C. Zhang et al., "Faster segment anything: Towards lightweight SAM for mobile applications," 2023, *arXiv:2306.14289*.

[21] Y. Xiong et al., "EfficientSAM: Leveraged masked image pretraining for efficient segment anything," 2023, *arXiv:2312.00863*.

[22] Z. Yang et al., "An easy zero-shot learning combination: Texture sensitive semantic segmentation IceHrNet and advanced style transfer learning strategy," 2023, *arXiv:2310.00310*.

[23] X. Zhang et al., "ICENET: A semantic segmentation deep network for river ice by fusing positional and channel-wise attentive features," *Remote Sens.*, vol. 12, no. 2, p. 221, Jan. 2020.

[24] A. Singh, H. Kalke, M. Loewen, and N. Ray, "River ice segmentation with deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7570–7579, Nov. 2020.

[25] Z. Wu, C. Shen, and A. van den Hengel, "Real-time semantic image segmentation via spatial sparsity," 2017, *arXiv:1712.00213*.

[26] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.

[27] X. Zhang et al., "ICENETv2: A fine-grained river ice semantic segmentation network based on UAV images," *Remote Sens.*, vol. 13, no. 4, p. 633, Feb. 2021.

[28] X. Zhang et al., "FastICENet: A real-time and accurate semantic segmentation model for aerial remote sensing river ice image," *Signal Process.*, vol. 212, Nov. 2023, Art. no. 109150.

[29] C. Pei, Y. She, and M. Loewen, "Deep learning based river surface ice quantification using a distant and oblique-viewed public camera," *Cold Regions Sci. Technol.*, vol. 206, Feb. 2023, Art. no. 103736.

[30] Z. Yang, J. Zong, Y. Zhu, X. Liu, R. Tao, and Y. Yu, "River ice regime recognition based on deep learning: Ice concentration, area, and velocity," *Water*, vol. 16, no. 1, p. 58, Dec. 2023.

[31] M. Temimi et al., "An automated system to monitor river ice conditions using Visible Infrared Imaging Radiometer Suite imagery," *Remote Sens.*, vol. 15, no. 20, p. 4896, Oct. 2023.

[32] P. Muhammad, C. Duguay, and K.-K. Kang, "Monitoring ice break-up on the Mackenzie River using MODIS data," *Cryosphere*, vol. 10, no. 2, pp. 569–584, Mar. 2016.

[33] G. M. Prakasam et al., "Detecting ice jams on the rivers in northern Finland using Sentinel-1," M.S. thesis, Inst. Atmos. Earth Syst. Res., Univ. Helsinki, Helsinki, Finland, 2022.

[34] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[35] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.

[36] Z. Xu, J. Geng, and W. Jiang, "MMT: Mixed-mask transformer for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613415.
[37] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601313.
[38] Y. Qiao et al., "SAM enhanced semantic segmentation for remote sensing imagery without additional training," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5610816.
[39] C. Surdu, C. Duguay, H. Pour, and L. Brown, "Ice freeze-up and break-up detection of shallow lakes in northern Alaska with spaceborne SAR," *Remote Sens.*, vol. 7, no. 5, pp. 6133–6159, May 2015.
[40] A. Beaton, R. Whaley, K. Corston, and F. Kenny, "Identifying historic river ice breakup timing using MODIS and Google Earth Engine in support of operational flood monitoring in northern Ontario," *Remote Sens. Environ.*, vol. 224, pp. 352–364, Apr. 2019.
[41] N. Chaouch, M. Temimi, P. Romanov, R. Cabrera, G. McKillop, and R. Khanbilvardi, "An automated algorithm for river ice monitoring over the Susquehanna River using the MODIS data," *Hydrological Processes*, vol. 28, no. 1, pp. 62–73, Jan. 2014.
[42] S. W. Cooley and T. M. Pavelsky, "Spatial and temporal patterns in Arctic River ice breakup revealed by automated ice detection from MODIS imagery," *Remote Sens. Environ.*, vol. 175, pp. 310–322, Mar. 2016.
[43] P. T. Bharathi and P. Subashini, "Texture based color segmentation for infrared river ice images using K-means clustering," in *Proc. Int. Conf. Signal Process., Image Process. Pattern Recognit.*, Feb. 2013, pp. 298–302.
[44] M. Tom, U. Kälin, M. Sütterlin, E. Baltsavias, and K. Schindler, "Lake ice detection in low-resolution optical satellite images," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 4, pp. 279–286, May 2018.
[45] X. Yang, T. M. Pavelsky, and G. H. Allen, "The past and future of global river ice," *Nature*, vol. 577, no. 7788, pp. 69–73, Jan. 2020.
[46] D. Purnell et al., "Observations of river ice breakup using GNSS-IR, SAR, and machine learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5800613.
[47] S. Ansari, C. D. Rennie, S. P. Clark, and O. Seidou, "IceMaskNet: River ice detection and characterization using deep learning algorithms applied to aerial photography," *Cold Regions Sci. Technol.*, vol. 189, Sep. 2021, Art. no. 103324.
[48] B. Fu, X. Sun, S. Ma, X. Ma, and Z. Liu, "CSEU-Net: ConvNeXt-SE-U-Net for river ice floe segmentation using unmanned aerial vehicle grayscale remote sensing images," *J. Appl. Remote Sens.*, vol. 18, no. 4, Oct. 2024, Art. no. 046505.
[49] W. Zhao et al., "Research on semantic segmentation algorithm of high latitude urban river ice based on deep transfer learning," *Int. J. Remote Sens.*, vol. 45, no. 13, pp. 4278–4299, Jul. 2024.
[50] S. Ansari, C. D. Rennie, S. P. Clark, and O. Seidou, "River ice detection and classification using oblique shore-based photography," *Cold Regions Sci. Technol.*, vol. 228, Dec. 2024, Art. no. 104303.
[51] M. Ayyad et al., "RIce-Net: Integrating ground-based cameras and machine learning for automated river ice detection," *Environ. Model. Softw.*, vol. 190, May 2025, Art. no. 106454.
[52] W. Zhao et al., "Improved ice channel segmentation and width calculation using drone-based optical sensors," *IEEE Sensors J.*, vol. 23, no. 24, pp. 31447–31460, Dec. 2023.
[53] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.
[54] Z. Tian, T. He, C. Shen, and Y. Yan, "Decoders matter for semantic segmentation: Data-dependent decoding enables flexible feature aggregation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3126–3135.
[55] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.
[56] E. Romera, J. M. Alvarez, L. M. Bergasa, and R. Arroyo, "ERFNet: Efficient residual factorized ConvNet for real-time semantic segmentation," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 1, pp. 263–272, Jan. 2018.
[57] H. Zhao et al., "PSANet: Point-wise spatial attention network for scene parsing," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 267–283.
[58] M. Yang, K. Yu, C. Zhang, Z. Li, and K. Yang, "DenseASPP for semantic segmentation in street scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3684–3692.
[59] H. Zhang et al., "Context encoding for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Mar. 2018, pp. 7151–7160.
[60] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5686–5696.
[61] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.
[62] A. Kirillov, R. Girshick, K. He, and P. Dollár, "Panoptic feature pyramid networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6399–6408.
[63] J. He, Z. Deng, L. Zhou, Y. Wang, and Y. Qiao, "Adaptive pyramid context network for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7519–7528.
[64] Z. Zhu, M. Xu, S. Bai, T. Huang, and X. Bai, "Asymmetric non-local neural networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 593–602.
[65] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.
[66] Y. Yuan, X. Chen, and J. Wang, "Object-contextual representations for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Springer, 2020, pp. 173–190.
[67] Y. Hong, H. Pan, W. Sun, and Y. Jia, "Deep dual-resolution networks for real-time and accurate semantic segmentation of road scenes," 2021, *arXiv:2101.06085*.
[68] W. Zhang, J. Pang, K. Chen, and C. C. Loy, "K-Net: Towards unified image segmentation," in *Proc. NIPS*, 2021, pp. 10326–10338.
[69] C. Yu, C. Gao, J. Wang, G. Yu, C. Shen, and N. Sang, "BiSeNet v2: Bilateral network with guided aggregation for real-time semantic segmentation," *Int. J. Comput. Vis.*, vol. 129, no. 11, pp. 3051–3068, Nov. 2021.
[70] M. Fan et al., "Rethinking BiSeNet for real-time semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9716–9725.
[71] M.-H. Guo, C.-Z. Lu, Q. Hou, Z. Liu, M. Cheng, and S. Hu, "SegNeXt: Rethinking convolutional attention design for semantic segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 1140–1156.
[72] R. Ranftl, A. Bochkovskiy, and V. Koltun, "Vision transformers for dense prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 12179–12188.
[73] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7262–7272.
[74] X. Chu et al., "Twins: Revisiting the design of spatial attention in vision transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 9355–9366.
[75] A. Paszke, A. Chaurasia, S. Kim, and E. Culurciello, "ENet: A deep neural network architecture for real-time semantic segmentation," 2016, *arXiv:1606.02147*.
[76] T. Wu, S. Tang, R. Zhang, J. Cao, and Y. Zhang, "CGNet: A light-weight context guided network for semantic segmentation," *IEEE Trans. Image Process.*, vol. 30, pp. 1169–1179, 2021.
[77] G. Li, I. Yun, J. Kim, and J. Kim, "DABNet: Depth-wise asymmetric bottleneck for real-time semantic segmentation," 2019, *arXiv:1907.11357*.
[78] M. Liu and H. Yin, "Feature pyramid encoding network for real-time semantic segmentation," 2019, *arXiv:1909.08599*.
[79] X. Zhang, Z. Chen, Q. M. J. Wu, L. Cai, D. Lu, and X. Li, "Fast semantic segmentation for scene perception," *IEEE Trans. Ind. Informat.*, vol. 15, no. 2, pp. 1183–1192, Feb. 2019.
[80] Y. Wang et al., "Lednet: A lightweight encoder–decoder network for real-time semantic segmentation," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 1860–1864.
[81] W. Han et al., "ContextNet: Improving convolutional neural networks for automatic speech recognition with global context," 2020, *arXiv:2005.03191*.
[82] R. P. K. Poudel, S. Liwicki, and R. Cipolla, "Fast-SCNN: Fast semantic segmentation network," 2019, *arXiv:1902.04502*.
[83] A. Chaurasia and E. Culurciello, "LinkNet: Exploiting encoder representations for efficient semantic segmentation," in *Proc. IEEE Vis. Commun. Image Process. (VCIP)*, Dec. 2017, pp. 1–4.
[84] J. Peng et al., "PP-LiteSeg: A superior real-time semantic segmentation model," 2022, *arXiv:2204.02681*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:08:38 UTC from IEEE Xplore. Restrictions apply.